

| Name    | IPv4 Address  |
|---------|---------------|
| PC-A1   | 192.168.10.10 |
| PC-A2   | 192.168.10.20 |
| PC-B1   | 192.168.20.10 |
| PC-B2   | 192.168.20.20 |
| VLAN 10 | 192.168.10.1  |
| VLAN 20 | 192.168.20.1  |

Rough Diagram:

VLAN 10 (PC-A1,PC-A1) to S1

VLAN 20 (PC-B1,PC-B2) to S1

S1 to R1

Steps:

- Physically connect the endpoints (PC-A1,PC-A2,PC-B1,PC-B2) to the switch with fastEthernet.
- Physically connect the switch to the router (R1) with gigabitEthernet.
- Assign static IPv4 Addresses to the endpoints through IP Configuration on the desktops.
- Enter router (R1) CLI, run following commands:
  - No shutdown gigabitEthernet 0/0/0
    - enable
    - configure terminal
    - interface gigabitEthernet 0/0/0
    - no shutdown
    - exit
  - IP Address to gigabitEthernet 0/0/0.10 (VLAN 10 Default Gateway)
    - interface gigabitEthernet 0/0/0.10
    - encapsulation dot1Q 10
    - ip address 192.168.10.1 255.255.255.0
    - exit
  - IP Address to gigabitEthernet 0/0/0.20 (VLAN 20 Default Gateway)
    - interface gigabitEthernet 0/0/0.20
    - encapsulation dot1Q 20
    - ip address 192.168.20.1 255.255.255.0
    - exit

- Save and Exit
  - end
  - write memory
- Enter switch (S1) CLI, run following commands:
  - NameVLAN 10
    - enable
    - configure terminal
    - vlan 10
    - name DEPARTMENT\_A
    - exit
  - Name VLAN 20
    - vlan 20
    - name DEPARTMENT\_B
    - exit
  - Configure connection to PC-A1 and associate to VLAN 10
    - interface fastEthernet 0/1
    - switchport mode access
    - switchport access vlan 10
    - no shutdown
    - exit
  - Configure connection to PC-A2 and associate to VLAN 10
    - interface fastEthernet 0/2
    - switchport mode access
    - switchport access vlan 10
    - no shutdown
    - exit
  - Configure connection to PC-B1 and associate to VLAN 20
    - interface fastEthernet 0/3
    - switchport mode access
    - switchport access vlan 20
    - no shutdown
    - exit
  - Configure connection to PC-B2 and associate to VLAN 20
    - interface fastEthernet 0/4
    - switchport mode access
    - switchport access vlan 20
    - no shutdown
    - exit
  - Configure connection to Router

- interface gigabitEthernet 0/1
- switchport mode trunk
- switchport trunk allowed vlan 10,20
- no shutdown
- exit
- Save and Exit
  - end
  - write memory

| Connectivity Tests |         |        |                                          |
|--------------------|---------|--------|------------------------------------------|
| Start              | End     | Result | Description                              |
| PC-A1              | PC-A2   | PASS   | VLAN ping tests                          |
| PC-B1              | PC-B2   | PASS   |                                          |
| PC-A1              | VLAN 10 | PASS   | Intra-VLAN to default gateway ping tests |
| PC-A2              | VLAN 10 | PASS   |                                          |
| PC-B1              | VLAN 20 | PASS   |                                          |
| PC-B2              | VLAN 20 | PASS   |                                          |
| PC-A1              | PC-B1   | PASS   | Cross-VLAN ping tests                    |
| PC-A2              | PC-B2   | PASS   |                                          |
| PC-A1              | PC-B2   | PASS   |                                          |
| PC-B1              | PC-A2   | PASS   |                                          |

| show vlan brief |         |        |                                                                                                                                                                   |
|-----------------|---------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VLAN            | Name    | Status | Ports                                                                                                                                                             |
| 1               | default | active | Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/2 |

|      |                    |        |              |
|------|--------------------|--------|--------------|
|      |                    |        |              |
| 10   | DEPARTMENT_A       | active | Fa0/1, Fa0/2 |
| 20   | DEPARTMENT_B       | active | Fa0/3, Fa0/4 |
| 1002 | fddi-default       | active |              |
| 1003 | token-ring-default | active |              |
| 1004 | fddinet-default    | active |              |
| 1005 | trnet-default      | active |              |

| show interfaces trunk |      |               |          |             |                               |
|-----------------------|------|---------------|----------|-------------|-------------------------------|
| Port                  | Mode | Encapsulation | Status   | Native vlan | Vlans allowed on trunk and... |
| Gig0/1                | on   | 802.1q        | trunking | 1           | 10,20                         |

| show interface status |      |            |      |        |       |              |
|-----------------------|------|------------|------|--------|-------|--------------|
| Port                  | Name | Status     | Vlan | Duplex | Speed | Type         |
| Fa0/1                 |      | connected  | 10   | a-full | a-100 | 10/100BaseTX |
| Fa0/2                 |      | connected  | 10   | a-full | a-100 | 10/100BaseTX |
| Fa0/3                 |      | connected  | 20   | a-full | a-100 | 10/100BaseTX |
| Fa0/4                 |      | connected  | 20   | a-full | a-100 | 10/100BaseTX |
| Fa0/5                 |      | notconnect | 1    | auto   | auto  | 10/100BaseTX |
| Fa0/6                 |      | notconnect | 1    | auto   | auto  | 10/100BaseTX |
| Fa0/7                 |      | notconnect | 1    | auto   | auto  | 10/100BaseTX |
| Fa0/8                 |      | notconnect | 1    | auto   | auto  | 10/100BaseTX |

|        |  |            |   |      |      |              |
|--------|--|------------|---|------|------|--------------|
| Fa0/9  |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/10 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/11 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/12 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/13 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/14 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/15 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/16 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/17 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/18 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/19 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/20 |  | notconnect | 1 | auto | auto | 10/100BaseTX |
| Fa0/21 |  | notconnect | 1 | auto | auto | 10/100BaseTX |

| Mac Address Table |                |         |        |
|-------------------|----------------|---------|--------|
| Vlan              | Mac Address    | Type    | Ports  |
| 1                 | 000a.f30d.ae01 | DYNAMIC | Gig0/1 |
| 10                | 000a.f30d.ae01 | DYNAMIC | Gig0/1 |
| 20                | 000a.f30d.ae01 | DYNAMIC | Gig0/1 |

| show ip interface brief |            |     |        |        |          |
|-------------------------|------------|-----|--------|--------|----------|
| Interface               | IP-Address | OK? | Method | Status | Protocol |
| GigabitEthernet0/0/0    | unassigned | YES | unset  | up     | up       |

|                         |              |     |        |                       |      |
|-------------------------|--------------|-----|--------|-----------------------|------|
| GigabitEthernet0/0/0.10 | 192.168.10.1 | YES | manual | up                    | up   |
| GigabitEthernet0/0/0.20 | 192.168.20.1 | YES | manual | up                    | up   |
| GigabitEthernet0/0/1    | unassigned   | YES | unset  | administratively down | down |
| GigabitEthernet0/0/2    | unassigned   | YES | unset  | Administratively down | down |